

Effects of Parameter Changes on Graphs of Quadratic Functions

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The **vertex form** of a quadratic function is

$$y = a(x - h)^2 + k$$

Vertex and Axis of Symmetry

The **vertex** of the parabola is (h, k) .

The **axis of symmetry** is the vertical line

$$x = h$$

The Parameter a

The parameter a determines the **opening direction** and the **width** of the parabola.

- ▶ If $a > 0$, the parabola opens upward.
- ▶ If $a < 0$, the parabola opens downward.
- ▶ If $|a| > 1$, the parabola becomes narrower.
- ▶ If $0 < |a| < 1$, the parabola becomes wider.

The Parameter h

The parameter h moves the graph horizontally.

- ▶ Positive h shifts the graph to the **right**.
- ▶ Negative h shifts the graph to the **left**.

The Parameter k

The parameter k moves the graph vertically.

- ▶ Positive k shifts the graph **upward**.
- ▶ Negative k shifts the graph **downward**.

Examples

Example 1

Compare $y = x^2$ and $y = 3x^2$.

$$a = 1 \longrightarrow a = 3$$

Since $|3| > 1$, the graph becomes **narrower**.
Both still open upward.

Example 2

Compare $y = (x - 2)^2$ and $y = (x + 2)^2$.

$$h = 2 \longrightarrow h = -2$$

The graph of $y = (x + 2)^2$ is shifted 4 units to the **left** of $y = (x - 2)^2$.

Example 3

State the vertex, opening direction, and axis of symmetry of

$$y = 2(x - 1)^2 + 4.$$

$$a = 2, \quad h = 1, \quad k = 4$$

Vertex: $(1, 4)$. Opens **upward**. Axis of symmetry: $x = 1$.

Practice Exercises

Practice 1

Describe the effect of changing a from 1 to 3
in

$$y = a(x - 2)^2 - 1.$$

Practice 2

Compare the graphs of

$$y = (x - 1)^2 + 2$$

and

$$y = (x - 4)^2 + 2.$$

Describe the effect of changing h .

Practice 3

State the vertex, opening direction, and axis of symmetry of

$$y = -2(x - 3)^2 + 5.$$

Practice 4

Which parameter should be changed to make

$$y = (x + 4)^2 - 2$$

open downward without changing its vertex?

Write the new equation.

Practice 5

Determine the values of a , h , and k in

$$y = -\frac{3}{2}(x + 5)^2 - 4.$$

Then describe the graph.

Activities

Activity 1

Describe the effect of changing a from -1 to $-\frac{1}{3}$ in

$$y = a(x - 2)^2 - 4.$$

Activity 2

Compare the graphs of

$$y = (x - 5)^2 + 2$$

and

$$y = (x - 5)^2 - 3.$$

Describe the effect of changing k .

Activity 3

Which parameter should be changed to move the graph

$$y = (x + 2)^2 - 1$$

five units upward? Write the new equation.

Activity 4

Explain how the graph changes when

$$y = \frac{1}{4}(x + 1)^2 - 3$$

is changed to

$$y = 5(x + 1)^2 - 3.$$

Activity 5

A parabola has vertex $(-3, 5)$, opens downward, and is wider than $y = x^2$. Write one possible equation in vertex form.

Thank you!